

CE 524: Applied Hydroclimatology

Course Code	CE 524
Course Name	Applied Hydroclimatology
L-T-P-C	3-0-0-3
Prerequisites	None; however, familiarity with any computer programming language (e.g., Python, R, MATLAB, etc.) will be helpful during the course.
Intended for	PG and Ph.D. students
Mutual Exclusion	None
Approval	56th BoA (Jul 2024)

Course Contents

- **Introduction to hydroclimatology Climate change:** causes and impacts, Impact on global hydrologic cycle; Climate feedback, Tipping points, Water management in changing climate; Concept and scope of hydroclimatology. (4 Lectures)
- **Module II:** Observation Types of research data; Qualitative and quantitative data; Measurement of precipitation, temperature, snow, streamflow, snow water equivalent; Missing data handling techniques; Hydroclimatic measurement network in the Himalayan regions. (4 Lectures)
- **Module III:** Modeling Fundamental principles in hydroclimatic modeling; Types of climate models; Simple climate models, Simple energy balance climate models (e.g., zero- dimensional, single-layer, and one-dimensional); General circulation model (GCM); Hydrologic modeling; types of hydrological models; Components of hydrologic models; SWAT model; VIC model; HEC-HMS.
- **Module IV:** Hydroclimatic data processing and analysis Gridding observed data; Reanalysis data: Satellite-derived data; Radar data; Climate data; Paleoclimatology data; Climate indices, Overview of CMIP6; Downscaling and bias correction of climate data; Impact assessment studies. (8 Lectures)
- **Module V:** Impact on hydrology, agriculture, and ecosystem Climate change impact on Dams; Weather and climate metrics for agriculture, Heat stress indices; Impact of climate change on food security in India; Energy limitation versus moisture limitation; Ecosystem services; Himalayan ecosystem and impact of climate change. (6 Lectures)
- **Module VI:** Hydroclimatic extremes: Droughts and Floods Drought definition: meteorological droughts, hydrological droughts, agricultural drought, socioeconomic drought; Drought indicators: Theory of run; Severity; Duration; Intensity; Frequency analysis; Impact of climate change on droughts; (6 Lectures)
- **Module VII:** Flood types; Extreme precipitation indices; Flood analysis; Flood Frequency analysis; Design Flood; Flood hazard and damage; Glacial lake outburst flood (GLOF); Impact of climate change on floods; Flood risk management and floodplain management. (6 Lectures)

Text Books

1. Shelton, M. L. , Hydroclimatology: perspectives and applications. Cambridge University Press, 2009.
2. VenTe, C., Maidment, D. R., & Mays, L. W., Applied hydrology, 1988.

References

1. Beven, K. J., **Rainfall-runoff modelling: the primer**, John Wiley & Sons, 2011.
2. McGuffie, K., & Henderson-Sellers, A., **The climate modelling primer**, John Wiley & Sons, 2014.